

8.8.2 Computer Exercises

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

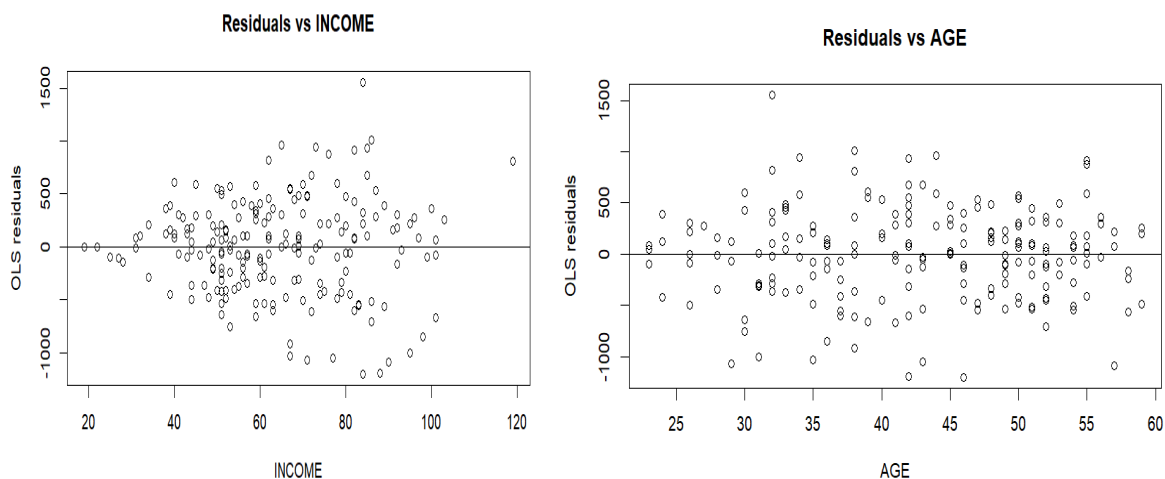
8.17 In this exercise, we explore the relationship between total household expenditures and expenditures on

a. The interval estimate is defined as:

$$[b_i \mp t_{\alpha/2} se(b_i)] \dots (1)$$

By running the code, we have the 95% confidence interval: [-135.3298 , -28.32302]

b. The residual plots are shown as below



We found that residual varies the same across ages, and the spread of the residual gets wider as income increases.

c. $H_0: \sigma_{small90}^2 = \sigma_{large90}^2$ v.s. $H_1: \sigma_{small90}^2 \neq \sigma_{large90}^2$ with $\alpha = 0.05$

We found that the test statistic is $\frac{\widehat{\sigma_{small90}^2}}{\widehat{\sigma_{large90}^2}} = 0.3221587$, and the rejection region is $\{F:$

$F < 0.6534355$ or $F > 1.530373\}$. Since the test statistic is in the critical region, we can

conclude that the variance of the 2 groups are not the same. The result is also consistent with (b).

d. The interval estimate is defined in (1), replacing $se(b_i)$ with robust standard error of b_i , and we have the result $[-139.323, -24.32986]$, we found that the confidence interval is wider than (a), to obtain more reliable inference under heteroskedasticity

e. Because $Var(e_i) = \sigma^2 \text{income}_i^2$, dividing the model by income_i makes the transformed error homoskedastic, allowing GLS estimation. By calculation in (a) and (d), we have the following confidence interval:

Conventional GLS: $[-119.8945, -33.71808]$

Robust GLS: $[-121.4134, -32.19919]$

Compare to (a), both conventional and robust GLS have narrower confidence interval, satisfying that GLS is have more precise result and narrower confidence interval.